CRIME INVESTIATION MANAGEMENT SYSTEM

Keerti Hosamani
[1] MCA Student, Department of Computer Applications, [BEC], Bagalkot, Karnataka, India
USN: 2BA24MC014

1. Abstract : Nowadays records are maintained on paper or in different systems, crime investigation is frequently slow and challenging. This study presents the Crime Investigation Management and Reporting System (CIMRS), a web-based tool that assists law enforcement in digitally managing investigations. FIRs, case information, evidence, tasks, and court data are all kept in one location by the system. To safeguard private information, it makes use of activity logs, role-based access, and secure login. A digital chain of custody and SHA256 hashing are used to guarantee evidence integrity. CIMRS increases the speed, security, and transparency of investigations by cutting down on paperwork and enhancing coordination**.**

Keywords : Crime Investigation, Case Management System, Digital Evidence ,Web-Based System, Information Security, Chain of Custody.

2. INTRODUCTION

Case handling is slow and ineffective because many areas still rely on manual records and dispersed digital files for crime investigation. FIRs, case diaries, evidence records, and court reports are examples of important documents that are frequently kept in physical files, which can cause data loss, duplication, and make it challenging to monitor the progress of a case. The quality of investigations and the administration of justice are impacted by these conventional methods' inability to deliver timely access to accurate information as the number of crimes rises.

A centralized, secure digital system to handle crime-related data is desperately needed given the speed at which information technology is developing. Law enforcement organizations can effectively record, store, and retrieve information with the aid of a computerized crime investigation system. Additionally, it facilitates better coordination between law enforcement and judicial authorities, decreases paperwork, and increases transparency.

A Crime Investigation Management System (CIMS) that digitizes the entire investigation process is proposed in this study. Through a centralized database, the system offers safe handling of FIRs, accused records, evidence details, and investigation updates. Sensitive data can only be accessed by authorized users thanks to role-based access control. The suggested system seeks to increase the effectiveness, accuracy, and accountability of crime investigation by automating investigation workflows and keeping organized digital records.

3. Overview

A Case Investigation Management System (CIMS) is a digital platform designed to help police manage criminal cases in an organized way. It stores important information such as FIR details, case records, accused information, evidence data, and court updates in one system. This reduces the need for physical files and registers. Officers can easily access case details whenever needed, which improves efficiency and saves time.

The system also supports multiple users like police officers, investigators, and supervisors. Each user can access only the information related to their role. This makes the system secure and ensures sensitive information is protected.

4. Problem Context

The existing manual case management system has many limitations. Paper files can be lost,

damaged, or misplaced. Searching for old case records takes a lot of time and effort. There is no proper tracking of evidence, which can create problems during court proceedings. Officers also struggle to monitor investigation progress, leading to delays in case completion.

Another major issue is the lack of accountability. Since actions are not recorded digitally, it is difficult to know who updated a case or handled evidence. This can cause confusion, mistakes, and legal challenges. Overall, the manual system increases workload and reduces investigation efficiency.

5. Solution Overview

- **24×7 System Access** for authorized police officers
- **Centralized Case Database** for all FIRs and investigations
- **Real-time Case Status Updates**
- **Secure User Login** with role-based permissions
- **Digital Evidence Management** with proper tracking
- **Case History and Timeline View**
- **Court Date and Hearing Tracking**
- **Automatic Activity Logs** for accountability
- **Quick Search** for cases, FIRs, and evidence
- **Report Generation** for case summaries and records
- **Reduced Paperwork** and manual effort
- **Improved Data Security** and privacy.

6. PROJECT OBJECTIVE

This project's primary goal is to create a Crime Investigation Management System (CIMS) that will replace the conventional manual crime record system with a safe and effective online platform. Due to the current system's reliance on paper files and disjointed records, case handling is laborious, prone to mistakes, and challenging to administer. The goal of this project is to consolidate crime-related data, including FIRs, case details, accused records, and investigation progress, into a single online system that authorized personnel can easily access and manage.

Enhancing data security, accuracy, and transparency in crime investigation procedures is another crucial goal. The system is made to minimize redundant data, guard against record loss, and enable quicker information retrieval. Sensitive data can only be viewed or updated by authorized users thanks to role-based access control. Time-stamping and automated logging support appropriate case activity monitoring and help ensure accountability during investigations.

Additionally, the project uses Python (backend), MySQL 8.0 (database), and HTML, CSS, JavaScript, and Bootstrap (frontend) to create a scalable and user-friendly web application. The system is designed to be user-friendly so that police personnel can use it without any technical issues.

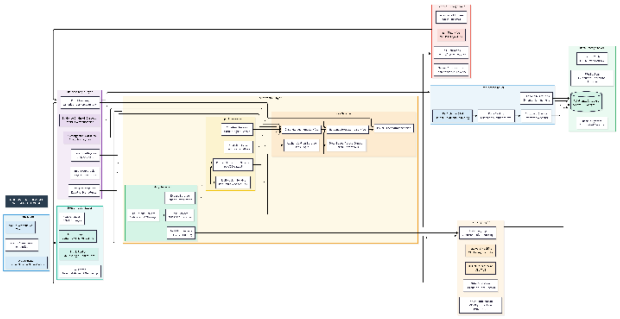
The suggested system lowers workload, saves time, and facilitates efficient decision-making by digitizing investigation workflows and reports, ultimately assisting law enforcement organizations in increasing the overall effectiveness of crime investigations.

7. PROPOSED SYSTEM

To address the shortcomings of the current manual system, a web-based Crime Investigation Management System (CIMS) is being proposed. It offers a centralised platform for the digital storage of crime details, FIRs, accused records, evidence, and case updates in a structured database.

7.1System Architecture Diagram

illustrates the overall system architecture of the proposed Crime Investigation Management System, showing the interaction between users, web application, backend services, and database.



7.2 DATA FLOW DIAGRAM (DFD)

Figure 1 illustrates the Level-0 and Level-1 Data Flow Diagrams of the system, showing the interaction between users, FIR processing, case management, evidence handling, and court modules.

Fig. 1.Data Flow Diagram of CIS **7.3 7.3 7.3**

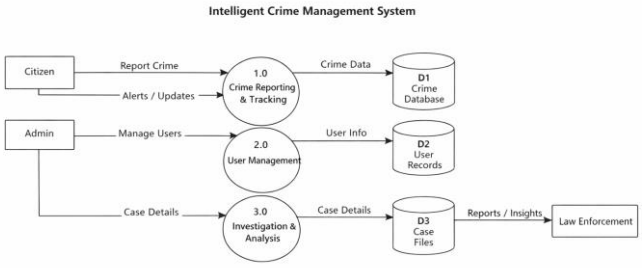


Fig. 1. Data Flow Diagram for *Intelligent Crime Management System.*

RELATIONSHIP DIAGRAM (ER)

The ER diagram represents relationships among core entities such as Users, FIRs, Cases, Evidence, Criminals, Victims, Witnesses, and Court Hearings.

Fig. 2. Entity Relationship Diagram of CIS

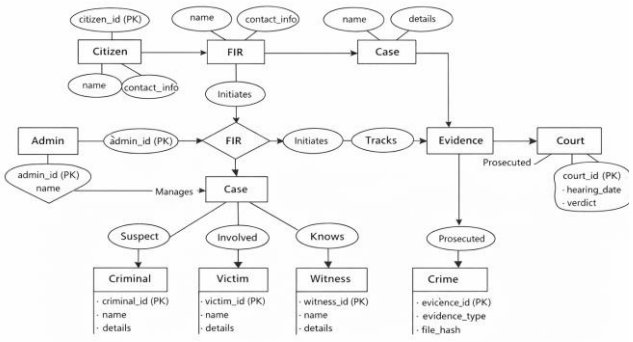


Fig. 2. Entity Relationship Diagram for Criminal Investigation System.

UML DIAGRAMS

UML diagrams including Use Case, Class, and Sequence diagrams are used to model system behavior and structure.

Fig. 3. UML Use Case Diagram

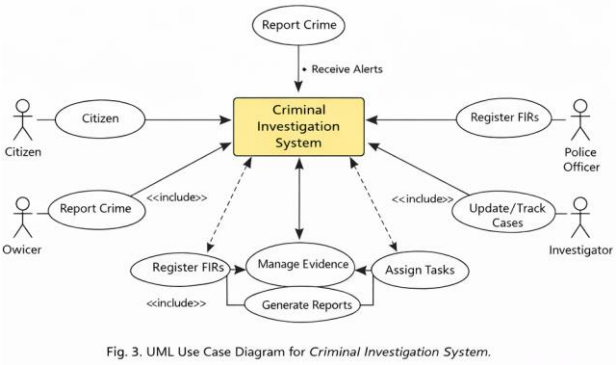


Fig. 3. UML Use Case Diagram for *Criminal Investigation System.*

Figure 7.1

The Case Management System is built in a simple and organized way so that all parts of the system work smoothly together.

First, users like police officers, investigators, and admins use the system through a web browser. They log in using their username and password. What they can see or do depends on their role.

Next, the user interface shows forms and pages where users can enter FIR details, update case progress, upload evidence, assign tasks, and check court dates. This makes the system easy to use.

Behind the scenes, the main system logic processes all actions. It checks user permissions, saves data correctly, updates case status, and keeps records of all actions done by users.

The system has a case module that manages all case information in one place. There is also an evidence module that safely stores evidence details and tracks who handled the evidence and when.

A court module keeps court case details and hearing dates. A task module helps assign work to officers and sends reminders.

All data is stored in a secure database, and a security layer ensures only authorized users can access sensitive information.

Overall, this design makes the system secure, easy to use, and able to handle many cases without confusion.

8. System Design

8.1 Class and Object Diagram

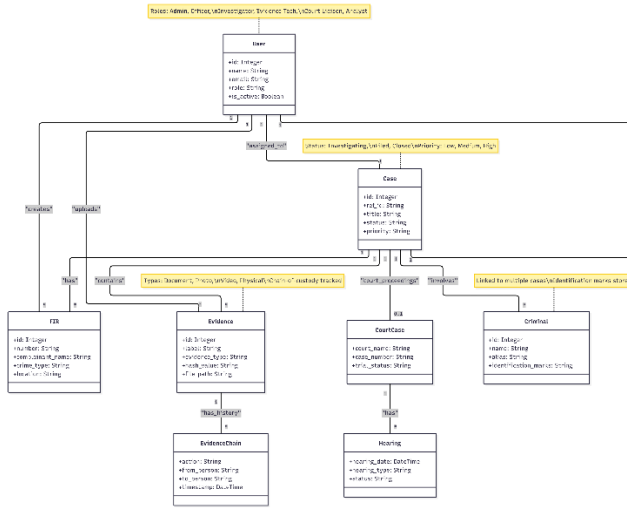


Figure 8.1

The Class and Object Diagram shows how different parts of the Case Management System are designed and how they are connected to each other. It explains the main classes (tables) in the system, their data, and how they work together to manage criminal cases.

User Class

The User class represents people who use the system, such as Admin, Police Officer, Investigator, Evidence Technician, Court Liaison, and Analyst. It stores user details like name, email, role, and login status. Users can create cases, update information, upload evidence, and perform assigned tasks based on their role.

Case Class

The Case class is the main class of the system. It contains case information such as case title, description, status, and priority. All other classes like Evidence, Criminal, CourtCase, and Task are connected to this class. This helps keep all case-related information in one place.

Evidence Class

The Evidence class stores information about evidence collected during an investigation. This includes documents, photos, videos, or physical items. Each evidence record is linked to a case and helps investigators track what evidence is available.

Evidence Chain Class

The Evidence Chain class keeps track of who handled the evidence and when. It records actions like collecting, transferring, or examining evidence. This helps maintain a proper chain of custody.

Criminal Class

The Criminal class stores details of suspects or accused persons. A criminal can be involved in more than one case, and a case can have multiple criminals. This helps track repeat offenders.

CourtCase Class

The CourtCase class contains court-related information such as court name, case number, and trial status. It links investigation work with legal proceedings.

Hearing Class

The Hearing class stores details of court hearings, including hearing date and status. Multiple hearings can be linked to one court case.

Task Class

The Task class is used to assign work to officers, such as evidence collection or witness questioning. It includes task priority and due date to ensure work is completed on time.

Object Diagram Explanation

The Object Diagram shows real examples of these classes in use. For example, one Case object may be linked to multiple Evidence objects, several Criminal objects, and many Task objects. This diagram helps understand how the system works in real-time with actual data.

8.2 Use Case Diagram



Figure 8.2

The Use Case Diagram of the proposed Case Investigation Management System (CIMS) illustrates the interaction between different user roles and the core functional modules of the system. It provides a high-level view of system behavior by identifying the actors involved and the services offered to each actor.

The primary actors in the system include Evidence Technician, Investigator, Admin, Officer, Analyst, Investigation Unit, Court Liaison, and General Users. Each actor is associated with specific use cases based on their roles and responsibilities within the investigation workflow.

The Evidence Technician interacts with the Evidence Management module, which enables secure uploading, verification, and maintenance of digital and physical evidence. This ensures proper chain-of-custody and evidence integrity throughout the investigation lifecycle.

The Investigator is responsible for Criminal Management and Case Management, including registering suspects, updating case details, assigning investigation tasks, and monitoring case progress. This role forms the core of the investigation process.

The Officer and Admin actors interact with the Case Management and Administration modules. The Admin has elevated privileges to manage users, roles, access permissions, and system configurations, ensuring secure and controlled system operation.

The Analyst supports the investigation process by accessing analytical data through the Administration module, enabling crime pattern analysis and decision support for ongoing cases.

The Investigation Unit collaborates across multiple cases, coordinating investigation activities and ensuring efficient workflow execution through the Case Management module.The Court Liaison interacts with the Court Proceedings module, which manages court dates, hearing details, legal documents, and case status updates, ensuring seamless integration between investigation and judicial.

The Court Liaison interacts with the system to manage legal documentation, prepare case summaries, submit evidence reports to the court, track court proceedings, and ensure that all judicial requirements are met in a timely and accurate manner.

Finally, General Users may have limited access to the system, such as submitting crime reports, viewing case updates (where permitted), or providing tips and information. Their interactions are controlled to ensure privacy, security, and data integrity.

Overall, the Use Case Diagram demonstrates how CIMS integrates multiple user roles and functional modules into a unified platform, ensuring transparency, accountability, collaboration, and effective management of case investigations from initiation to legal resolution.

- The use case diagram helps to understand how users interact with the system at a basic level.
- It shows the system boundary and separates user actions from internal system functions.
- Each user role is given limited access based on their responsibilities to improve security.
- Common functions are shared across roles to avoid duplication of work.
- The diagram supports teamwork by allowing multiple users to work on the same case.
- Error situations, such as invalid data or unauthorized access, are handled by the system.

9. EXISTING SYSTEM

The majority of the current crime investigation system is paper-based and manual. Police stations physically store registers and files that contain information about crimes, FIRs, and evidence. Because of this, handling records is slow, challenging, and vulnerable to loss or damage over time.

It takes a lot of time to search through previous records or monitor the status of cases because of the manual process.

It takes more work to update records because files can be lost, damaged, or duplicated. Police personnel have more work to do as a result, and reporting and investigation are delayed.

Project Module Formulation

To guarantee the methodical execution of investigation tasks and the safe handling of sensitive data, the proposed Crime Investigation Management System (CIS) is organised into clearly defined functional modules. Scalability, maintainability, and operational efficiency are all enhanced by the modular design. Module for Management.

A. User Authentication

Secure user authentication and authorisation are handled by this module. In order to limit system functions based on user responsibilities, such as administrator, investigator, evidence handler, and court liaison, it supports role-based access control.

B. Module for FIR Registration

The digital registration of First Information Reports (FIRs) is made easier by this module. It keeps track of crime classification, incident details, and complainant information. The main source of information used to start investigation cases is registered FIRs.

C. Module for Case Investigation

The creation, assignment, and monitoring of cases are handled by the case investigation module. To guarantee
organised and open investigation progress, it keeps track of investigation phases, case status, and officer assignments.

D. Module for Criminal Record Management

This module keeps thorough records of both known criminals and those who have been accused. It facilitates the linking of criminal profiles to several cases, allowing for efficient tracking and analysis throughout investigations.

E. Chain-of-Custody and Evidence Management

Module Documentation of both digital and tangible evidence is handled by this module. A structured chain-of-custody system documents evidence handling activities for legal compliance, and cryptographic verification ensures the integrity of the evidence.

F. Module for Tracking Investigation

Workflow Standardised investigation phases, including gathering evidence, questioning witnesses, and final reporting, are defined in this module. It permits real-time tracking of the investigation's progress and guarantees procedural consistency.

G. Module on Court and Legal Proceedings

Court case details, hearing schedules, applied legal sections, and judgements are all managed by this module. It improves communication between judicial authorities and investigative teams.

H. Module for Audit Logging and Monitoring

This module records every action taken by the system. In order to guarantee accountability, transparency, and traceability, it records user actions with timestamps.

10. Functional Requirements

The functional requirements specify the primary capabilities of the Criminal Investigation System.

User Authentication and Authorization:
The system shall provide secure user login using unique credentials and enforce role-based access control for administrators and investigation personnel.

FIR Registration and Management:

The system shall allow authorized users to register, update, and view First Information Reports in a structured digital format.

Case Management:
The system shall support creation, assignment, monitoring, and closure of investigation cases with proper status tracking.

Criminal Record Management:
The system shall maintain and manage criminal profiles, enabling association of individuals with multiple cases.

Evidence Management:
The system shall support secure documentation and tracking of physical and digital evidence, maintaining proper chain-of-custody records.

Court Proceedings Management:
The system shall store and manage court-related details including hearing schedules, legal sections, and judgments.

Search and Report Generation:
The system shall provide efficient search facilities and generate investigation and administrative reports.

Audit Logging:
The system shall record user activities to ensure accountability and traceability.

11. Non-Functional Requirements

The non-functional requirements define the performance and quality characteristics of the system.

Security:
The system shall ensure confidentiality and integrity of data through authentication, authorization, and controlled database access.

Performance:
The system shall provide acceptable response time for data processing and retrieval under normal operational conditions.

Scalability:
The system shall support future expansion in terms of users, data volume, and functionality.

Reliability:
The system shall operate consistently and preserve data integrity during continuous usage.

Usability:
The system shall provide a user-friendly interface that requires minimal training.

Maintainability:
The system shall be modular in design to support easy updates and enhancements.

Availability:
The system shall be accessible to authorized users with minimal downtime.

11. Future Enhancement

Artificial Intelligence Integration:
Future versions of the Case Investigation Management System can incorporate artificial intelligence and machine learning techniques to analyze crime data, identify crime patterns, and support predictive investigation. This enhancement can assist investigators in faster decision-making and proactive crime prevention.

Mobile Application Support:
The system can be extended with a mobile application to allow field officers to access case information, upload evidence, and update investigation details in real time. This will improve operational efficiency and reduce delays during field investigations.

Biometric Authentication:
To enhance system security, biometric authentication methods such as fingerprint or facial recognition can be integrated. This ensures stronger user authentication and prevents unauthorized access to sensitive investigation data.

Blockchain-Based Evidence Management:
Blockchain technology can be adopted to maintain tamper-proof digital evidence storage and secure chain-of-custody records. This enhancement increases the reliability and legal admissibility of digital evidence.

12. Conclusion

This paper presented a web-based Crime Investigation System designed to enhance the management of criminal investigation activities through digital transformation.
The proposed system replaces traditional manual methods by providing centralized management of FIRs, cases, criminal records, evidence, and court proceedings. Role-based access control and audit logging mechanisms ensure data security, accountability, and transparency throughout the investigation process.
The modular and scalable design supports efficient data handling and future system expansion. By improving accuracy, reducing manual workload, and enabling timely access to information, the Crime Investigation System contributes to more effective decision-making and supports modern law enforcement agencies in achieving reliable and efficient investigation outcomes.

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